

Unimodularity

G: finite abstract simplicial complex.

$C(x) = \{y \in G, y \subset x\}$ closed set

$U(x) = \{y \in G, x \subset y\}$ open set

$L(x, y) = \chi(C(x) \cap C(y))$
connection matrix
 $g(x, y) = \omega(x)\omega(y)\chi(U(x) \cap U(y))$
Green function

$D = d + d^*$ was based on incidence
 L is based on intersection
 D was never invertible L always is

$$\det(L) = \prod_{x \in G} \omega(x)$$

1
Unimodularity

$$L^{-1} = g$$

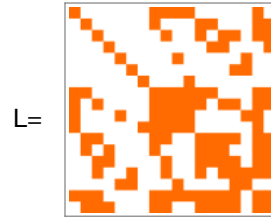
2
Green Star

$$\sum_{x,y} g(x, y) = \chi(G)$$

3
Energy Theorem

Example

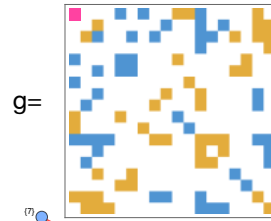
$G = \{\{1\}, \{2\}, \{3\}, \{4\}, \{5\}, \{6\}, \{7\},$
 $\{1,2\}, \{1,3\}, \{1,5\}, \{1,7\}, \{2,3\}, \{2,4\}, \{3,4\}, \{3,6\}, \{5,6\},$
 $\{1,2,3\}, \{2,3,4\}\}$ $f=(7, 9, 2)$ $b=(1,1,0)$



entries
of L are

1
0

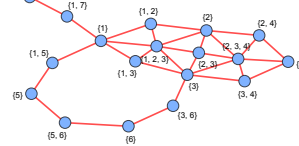
$\det(L) = -1$



entries
of g are

-2
1
-1
0

$\det(g) = -1$

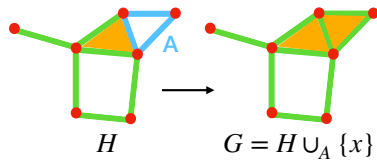


Proof of

1

$$\det(L) = \prod_{x \in G} \omega(x)$$

Build up complex inductively by
attaching k-simplex x to (k-1) sphere.



Right hand side: multiply by $\omega(x)$

Left hand side: define $\psi(H) = \det(L(H))$

$$Y: A \rightarrow (\psi(G \cup_A x) - \psi(G))$$

is a valuation satisfying $Y(A) = -\psi(A)$ for
complete A. Since $-\frac{Y(A)}{\psi(G)} = 1$ for
complete A, it must be $\chi(A)$. Therefore,
 $\psi(G \cup_A \{x\}) = \psi(G)(1 - \chi(A))$
 $= \psi(G)\omega(x)$ QED

Proof of

2

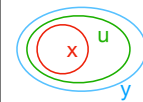
$$L^{-1} = g$$

$$L(u, v) = \sum_{x, x \subset u \cap v} h(x)$$

$$g(v, w) = \sum_{y, v \cup w \subset y} \omega(v)\omega(w)h(y)$$

$$\sum_v L(u, v)g(v, w) = \sum_v \sum_{x, x \subset u \cap v} \sum_{y, v \cup w \subset y} \omega(v)\omega(w)h(x)h(y)$$

Case $u=w$:
(diagonal)

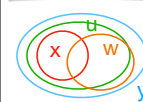


$$\sum_v [\sum_{x \subset u \cap v, v \cup w \subset y} h(x)\omega(v)\omega(w)h(y)]$$

$$= h(u)^2 = 1$$

(only $x=y=v=u$ produces a
non-zero situation)

Case $u \neq w$:
(off-diagonal)



$$\sum_v [\sum_{x \subset u \cap v, v \cup w \subset y} h(x)\omega(v)\omega(w)h(y)]$$

$$= 0$$

(as $x=y$ is now not possible
and terms cancel)

QED

Proof of

3

$$\sum_{x,y} g(x, y) = \chi(G)$$

$$g(x, x) = \chi(U(x)) = 1 - \chi(S(x))$$

Proof: Cramer gives

$$g(x, x) = \psi(G \setminus x) / \psi(G)$$

$$= (1 - \chi(S(x)))$$

We call the later

multiplicative Poincare-Hopf

Hyperbolicity: $S_f^-(x) \oplus S_f^+(x) = S(x)$

So: $i_f(x) i_{-f}(x) = (1 - \chi(U(x)))$

Take coloring $f = \dim$, where $i_f(x) = \omega(x)$

$$k(x) = \omega(x)(1 - \chi(S(x)))$$

is $i_{-f}(x)$ and so is a curvature.

Finally, equate the potential

$$V(x) = \sum_y g(x, y)$$

with $k(x)$.

QED

Why interesting?

Linear algebra

A) We can easily generate large
non-trivial integer matrices which
are unimodular.

Hearing Euler

B) The number of positive
eigenvalues minus the number of
negative eigenvalues is the Euler
characteristic.

Stable vaccum

C) We have a potential theoretic set-
up, where all Green functions are
bounded and have limited range.

Rich extensions

D) The situation links many topics
and generalizes vastly, for example
to k-point Green function set-ups.